

P10: Integration (Year 1)

1. Integration as Reverse Differentiation

Indefinite Integration (Antidifferentiation):

If $\frac{d}{dx}[F(x)] = f(x)$, then $\int f(x) \, dx = F(x) + C$

The constant C (constant of integration) is essential — it accounts for the fact that many functions share the same derivative. Always include C unless evaluating a definite integral.

Standard Results

Function $f(x)$	Integral $\int f(x) \, dx$	Condition
x^n	$\frac{x^{n+1}}{n+1} + C$	$n \neq -1$
kx^n	$\frac{kx^{n+1}}{n+1} + C$	$n \neq -1$
e^x	$e^x + C$	—
$\frac{1}{x}$	$\ln x + C$	$x \neq 0$
$\cos x$	$\sin x + C$	x in radians
$\sin x$	$-\cos x + C$	x in radians
k (constant)	$kx + C$	—

Power Rule for Integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

Mnemonic: "Raise the power by 1, divide by the new power."

Example — Integrate each:

$$(a) \int 3x^2 + 4x - 1 dx = x^3 + 2x^2 - x + C$$

$$(b) \int \frac{1}{x^3} dx = \int x^{-3} dx = \frac{x^{-2}}{-2} + C = -\frac{1}{2x^2} + C$$

$$(c) \int 5\sqrt{x} dx = \int 5x^{1/2} dx = 5 \cdot \frac{x^{3/2}}{3/2} + C = \frac{10}{3}x^{3/2} + C$$

$$(d) \int \frac{x^2 + 3}{x} dx = \int x + 3x^{-1} dx = \frac{x^2}{2} + 3 \ln |x| + C$$

Exam Tip: Always expand or simplify before integrating. You cannot integrate a product or quotient term-by-term without simplifying first (without more advanced techniques like integration by parts or substitution, which are Year 2).

2. Finding the Constant of Integration

When given a boundary condition (a point the curve passes through), substitute to find C .

Example — Given $\frac{dy}{dx} = 3x^2 - 2x$ **and** $y = 5$ **when** $x = 1$, **find** y :

$$y = \int 3x^2 - 2x dx = x^3 - x^2 + C$$

Substitute $x = 1, y = 5$: $5 = 1 - 1 + C \Rightarrow C = 5$

Therefore: $y = x^3 - x^2 + 5$

3. Definite Integrals

Definite Integral:

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

The constant C cancels, so it is not needed for definite integrals. The result is a number (not a function).

Example — Evaluate $\int_1^4 (3x^2 - 2x + 1) dx$:

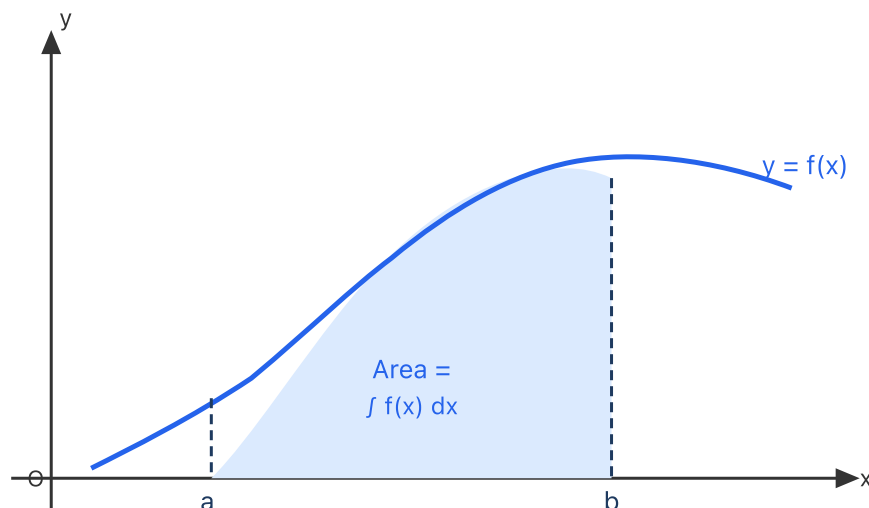
$$\begin{aligned} &= [x^3 - x^2 + x]_1^4 \\ &= (64 - 16 + 4) - (1 - 1 + 1) \\ &= 52 - 1 = 51 \end{aligned}$$

4. Area Under a Curve

The area between a curve $y = f(x)$ and the x-axis from $x = a$ to $x = b$ is:

$$A = \int_a^b f(x) dx$$

provided $f(x) \geq 0$ throughout the interval (curve is above the x-axis).



Areas Below the x-axis

If $f(x) < 0$ in an interval, the definite integral gives a **negative** value. The **area** is always positive.

Key Rule: To find the total area when a curve crosses the x-axis, split at the crossing points and take the **absolute value** of each section:

$$\text{Total area} = \left| \int_a^c f(x) dx \right| + \left| \int_c^b f(x) dx \right|$$

where c is where the curve crosses the x-axis. Never just evaluate $\int_a^b f(x) dx$ — sections below the axis will cancel sections above.

Example — Find the area bounded by $y = x(x - 2)$ and the x-axis:

Roots: $x = 0$ and $x = 2$. Curve is *below* x-axis for $0 \leq x \leq 2$.

$$\begin{aligned} \int_0^2 x(x - 2) dx &= \int_0^2 x^2 - 2x dx = \left[\frac{x^3}{3} - x^2 \right]_0^2 \\ &= \left(\frac{8}{3} - 4 \right) - 0 = -\frac{4}{3} \end{aligned}$$

$$\text{Area} = \left| -\frac{4}{3} \right| = \frac{4}{3} \text{ square units.}$$

5. Area Between Two Curves

The area enclosed between $y = f(x)$ (upper curve) and $y = g(x)$ (lower curve) from $x = a$ to $x = b$ is:

$$A = \int_a^b [f(x) - g(x)] dx$$

First find the limits by solving $f(x) = g(x)$ for the intersection points.

Example — Find the area between $y = x + 2$ and $y = x^2$:

Intersections: $x + 2 = x^2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0$

So $x = -1$ and $x = 2$. Line is above parabola in this interval.

$$A = \int_{-1}^2 [(x + 2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2$$

$$\text{At } x = 2: 2 + 4 - \frac{8}{3} = 6 - \frac{8}{3} = \frac{10}{3}$$

$$\text{At } x = -1: \frac{1}{2} - 2 + \frac{1}{3} = -\frac{7}{6}$$

$$A = \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{20}{6} + \frac{7}{6} = \frac{27}{6} = \frac{9}{2} \text{ square units.}$$

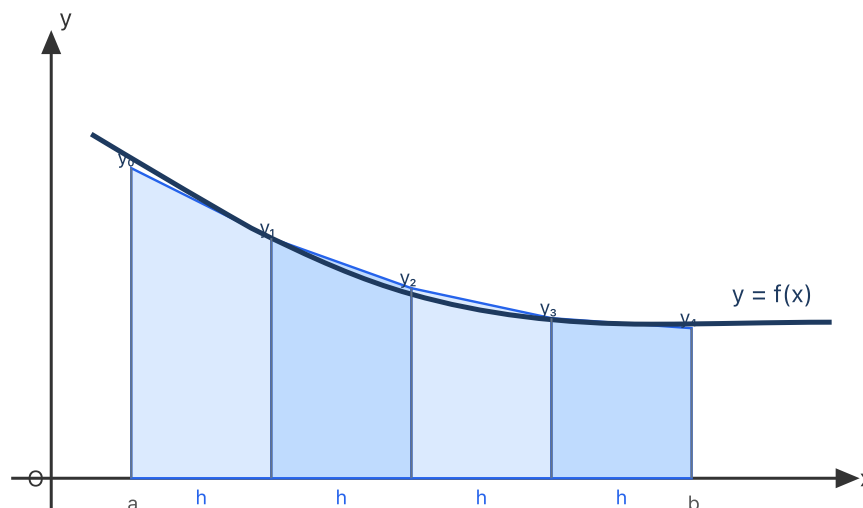
6. The Trapezium Rule

Trapezium Rule:

$$\int_a^b f(x) dx \approx \frac{h}{2} [y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n]$$

where $h = \frac{b-a}{n}$ is the width of each strip and $y_i = f(a + ih)$.

The rule approximates the area under the curve by replacing it with a series of trapezoids.



Example — Estimate $\int_1^3 \sqrt{x} \, dx$ using the trapezium rule with 4 strips:

$$h = \frac{3 - 1}{4} = 0.5$$

x -values: 1, 1.5, 2, 2.5, 3

y -values: $\sqrt{1} = 1$, $\sqrt{1.5} \approx 1.2247$, $\sqrt{2} \approx 1.4142$, $\sqrt{2.5} \approx 1.5811$, $\sqrt{3} \approx 1.7321$

$$\begin{aligned} &\approx \frac{0.5}{2} [1 + 2(1.2247 + 1.4142 + 1.5811) + 1.7321] \\ &= 0.25 [1 + 2(4.22) + 1.7321] \\ &= 0.25 [1 + 8.44 + 1.7321] = 0.25 \times 11.172 \approx 2.793 \end{aligned}$$

(Exact value: $\frac{2}{3}[x^{3/2}]_1^3 = \frac{2}{3}(3\sqrt{3} - 1) \approx 2.797$)

Trapezium Rule — Key Points:

- The formula is given in the Edexcel formula booklet — you must substitute correctly
- The first and last y -values (y_0 and y_n) are added once; all middle values are doubled
- For a curve that is **concave up** (like x^2), the trapezium rule gives an **overestimate**
- For a curve that is **concave down** (like $\sqrt{x}$), it gives an **underestimate**
- More strips = greater accuracy

Key Exam Points for P10:

- Indefinite integrals: always include $+C$; definite integrals: never include C
- Areas below x-axis give negative integrals — always take absolute value for area questions
- Split integrals at x-intercepts when curve crosses the x-axis within the limits
- For area between curves: always identify which curve is on top; sketch if unsure
- Trapezium rule: label your y -values clearly in a table to avoid errors